1. Purpose

Guide our production line employees to operate correctly and ensure product quality.

2. Scope of application

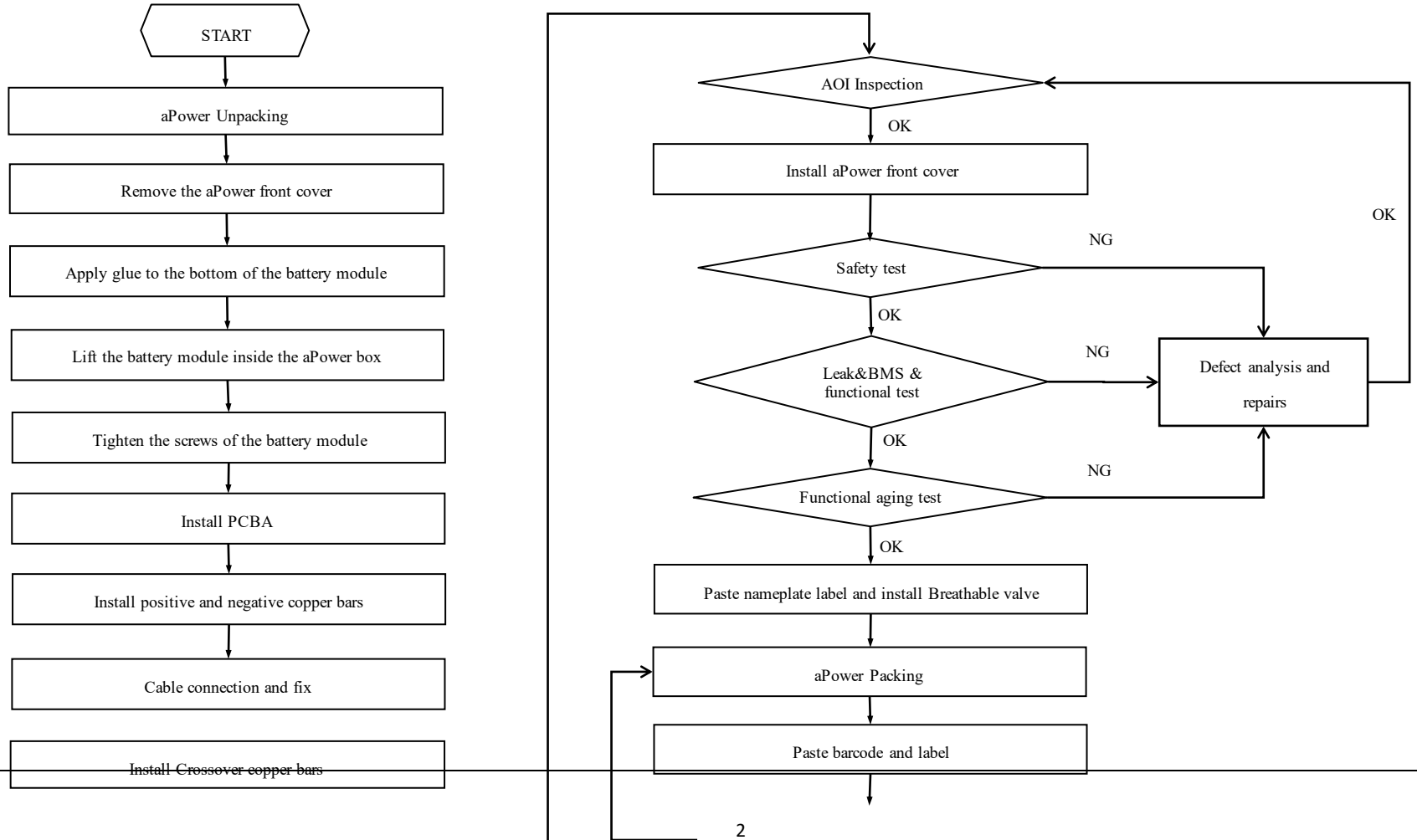
This process document is applicable to the assembly guidance of MIS_2.

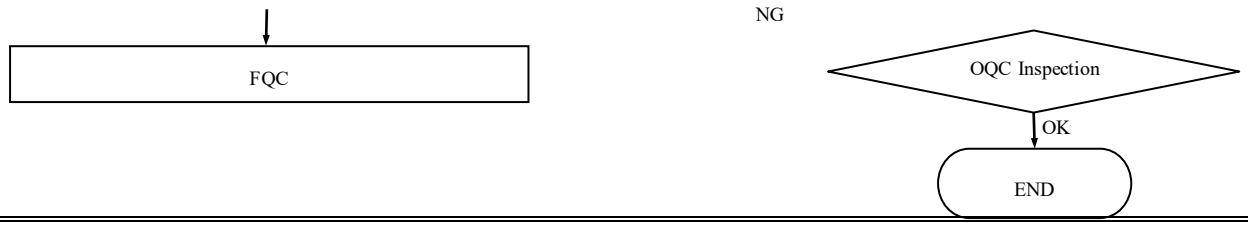
NO.	Code	Coded Description	Notes
1	10050018	Residential Energy Storage - aPower8000A - A100136F - NA - 1 - 10kW/15kWh - Standard Model - MIS_2	
2			
3			

3.Document Revision Record:

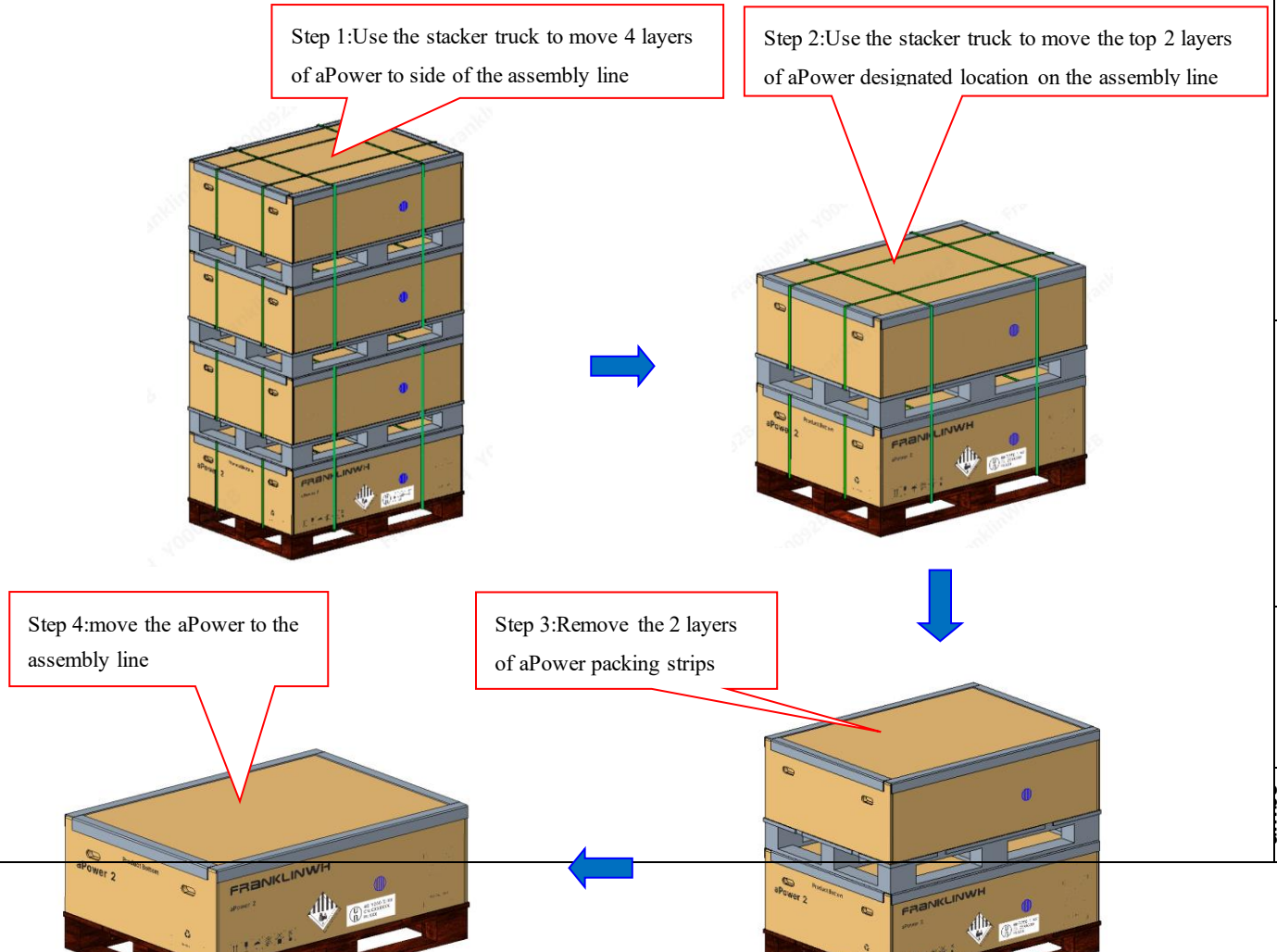
Version No.	Revision No.	Contents and reasons for revision	Revision time
1.0	PFarchive-20250314001	First Edition	14/03/2025
1.1	PFarchive-20250314001	Combine the two stations for installing copper bars Add shipping label check	04/14/2025

4.Process flow chart





Offline process1:aPower Unpacking



Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	10150003	aPower Bare machine	1
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Use the stacker truck to move 4 layers of aPower to side of the assembly line;
Step 2:Use the stacker truck to move the top 2 layers of aPower designated location on the assembly line;
Step 3:Remove the 2 layers of aPower packing strips;
Step 4: move the aPower to the assembly line;

Notes

and

NO.	Code	Description	Quantity
1	/	Diagonal pliers	1

2	/	Stacker truck	1
3	/	Cantilever crane	1
4			
5			

Process 1:aPower Unpacking

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Remove the top carton and place it on the circulate assembly line with robot;

Notes

Tool and equipment

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			

Process 2:Remove the aPower front cover

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			
6			

Procedure of operation

Step 1:.Use an assistive arm to Remove the 4pcs fixing screws on the aPower front cover;

Step 2: Use a suction cup to entire the aPower front cover to the circulate assembly line;

Notes

1.The removed screws are placed in the scrap box and no longer used.

Tool and equipment

NO.	Code	Description	Quantity
1	/	Assistive arm	1
2	/	CCD automatic screw equipment	1
3			
4			
5			

Process 3:Apply glue to the bottom of the battery module

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	27040015	Thermal conductive structural adhesive	540+/-20 g
2			
3			
4			
5			
6			

Procedure of operation

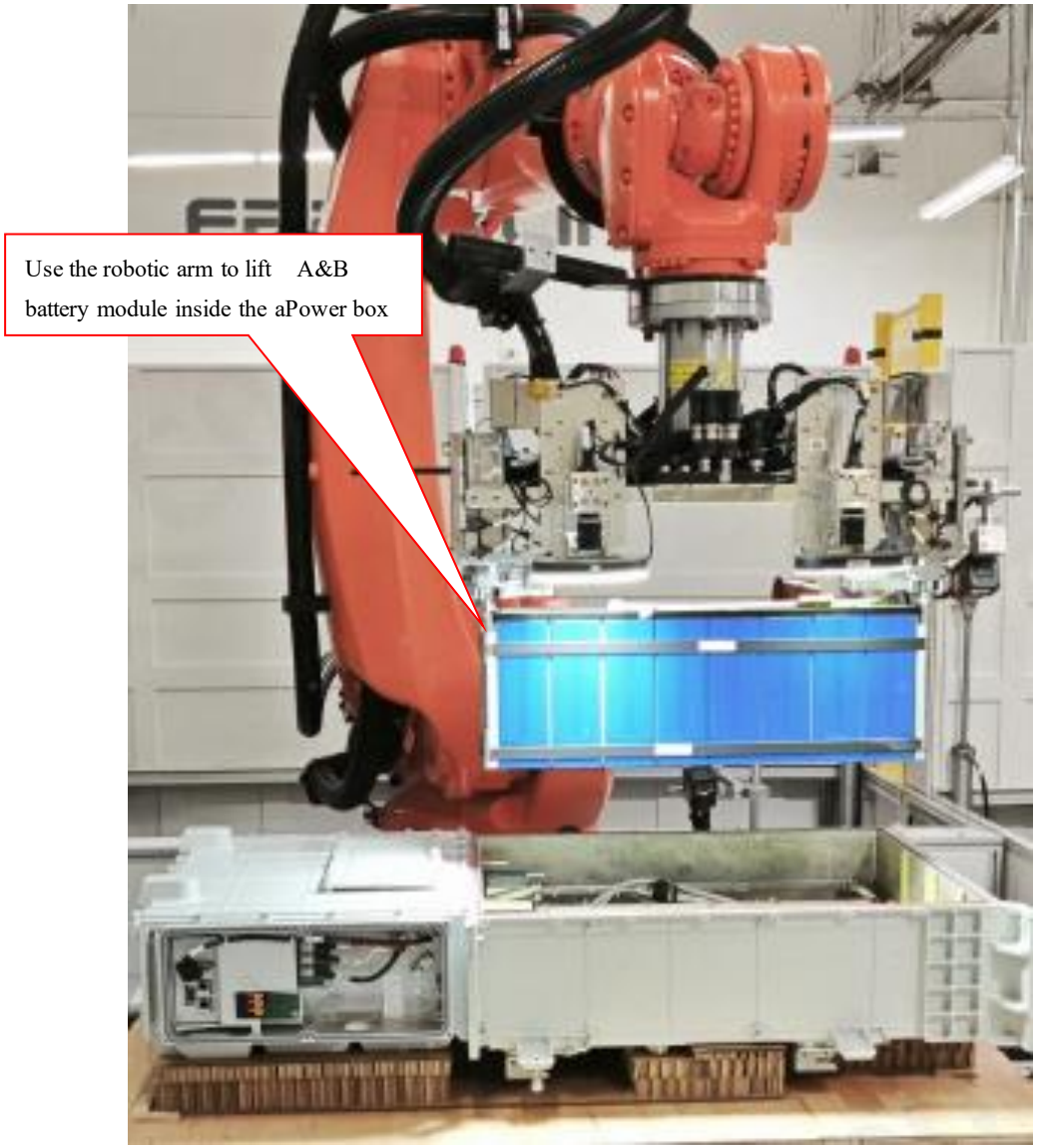
Step 1:Use an automatic gluing equipment to evenly apply thermal conductive structural glue at the position shown in the figure;

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Automatic gluing equipment	1
2			
3			
4			
5			

Process 4:Lift the battery module inside the aPower box



Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			
6			

Procedure of operation

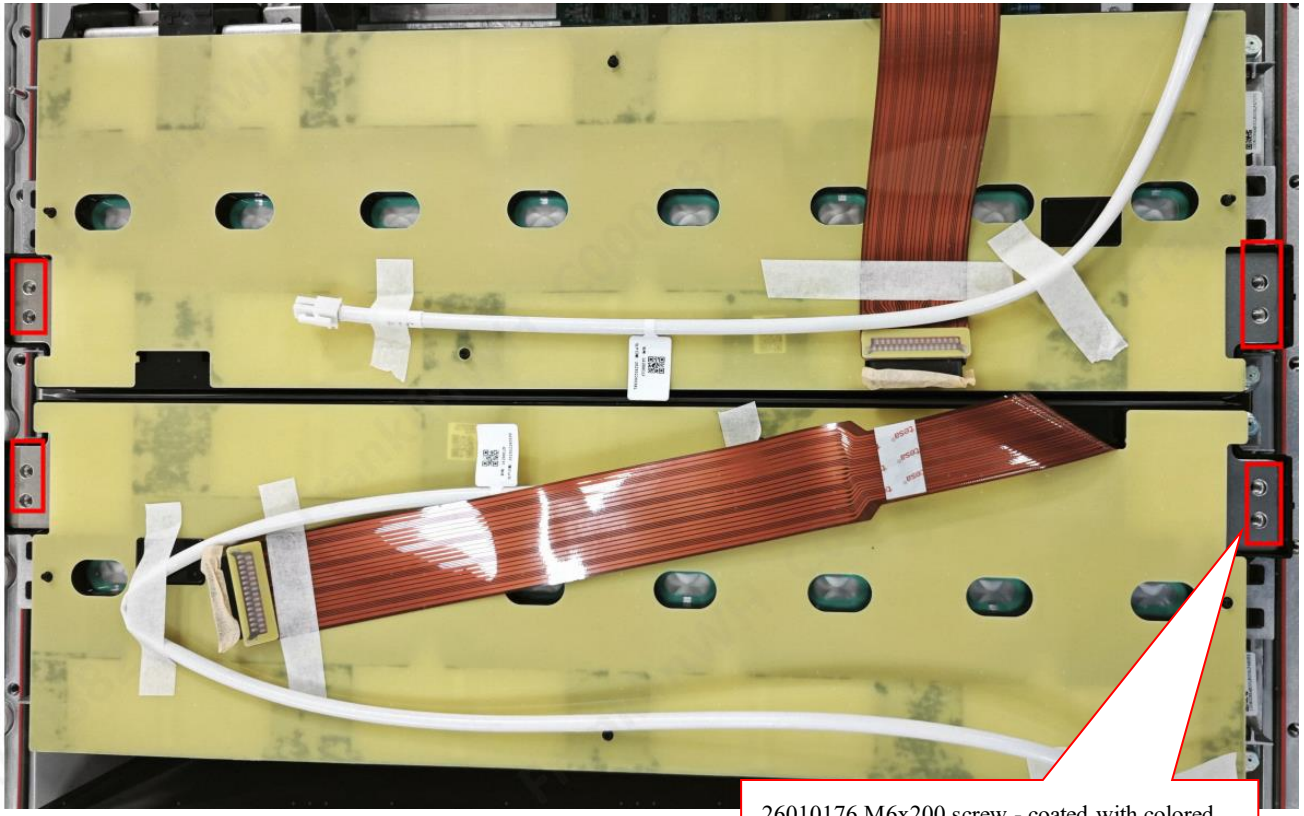
Step 1:Use the robotic arm to lift A&B battery module inside the aPower box;

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Robot	1
2			
3			
4			
5			

Process 5:tighten the screws of the battery module



26010176 M6x200 screw - coated with colored zinc 8pcs;Tightening torque 6N. m

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	26010176	M6x200 screw - coated with colored zinc	8
2			
3			
4			
5			
6			

Procedure of operation

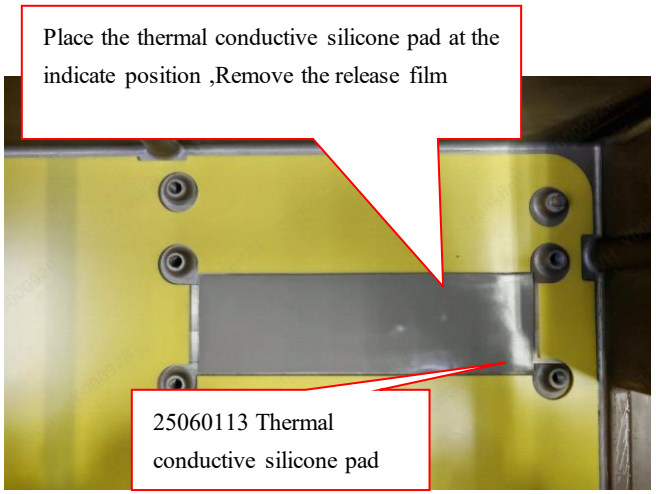
Step 1: tighten the 8pcs screws of the battery module;

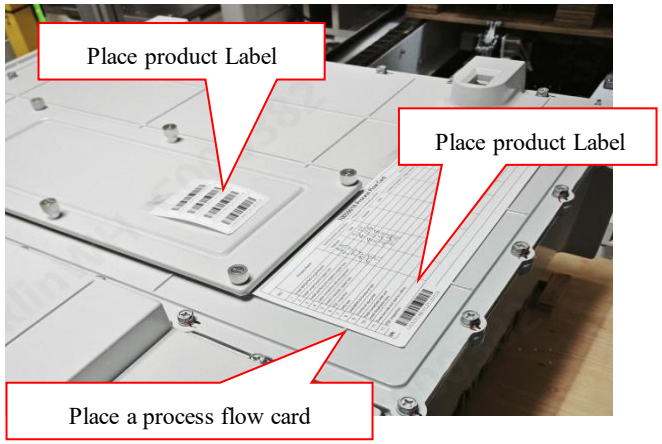
Notes

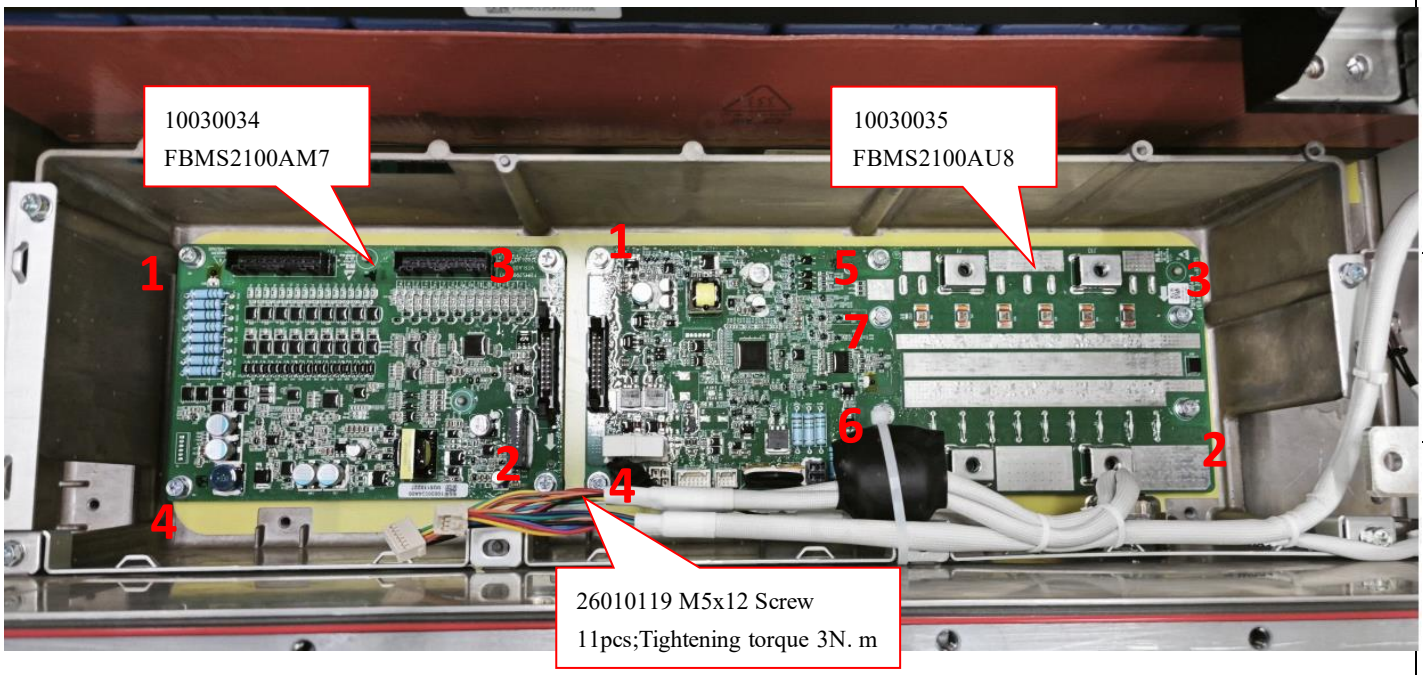
Tool and equipment

NO.	Code	Description	Quantity
1	/	robot	1
2			
3			

Process 6: Install PCBA







Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	25060113	Thermal conductive silicone pad	1
2	10030035	FBMS2100AU8	1
3	10030034	FBMS2100AM7	1
4	26010119	M5x12 combination Screw-coated with colored zinc	11

Procedure of operation

Step 1:Place the thermal conductive silicone pad at the indicate position

Step 2:Remove the release film of the thermal conductive silicone pad;

Step 3:Install FBMS2100AM7 and Tighten the 4 screws in the order shown in the diagram;

Step 4:Install FBMS2100AU8 and Tighten the 7 screws in the order shown in the diagram;

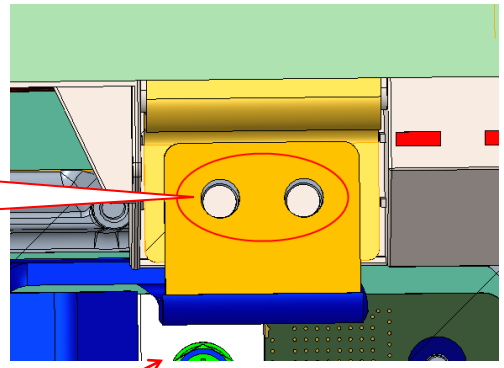
Step 5: Place a process flow card and stick 1pcs bar-code on the process flow car,stick 2pcs bar-code on the product

Notes

Tool and equipment

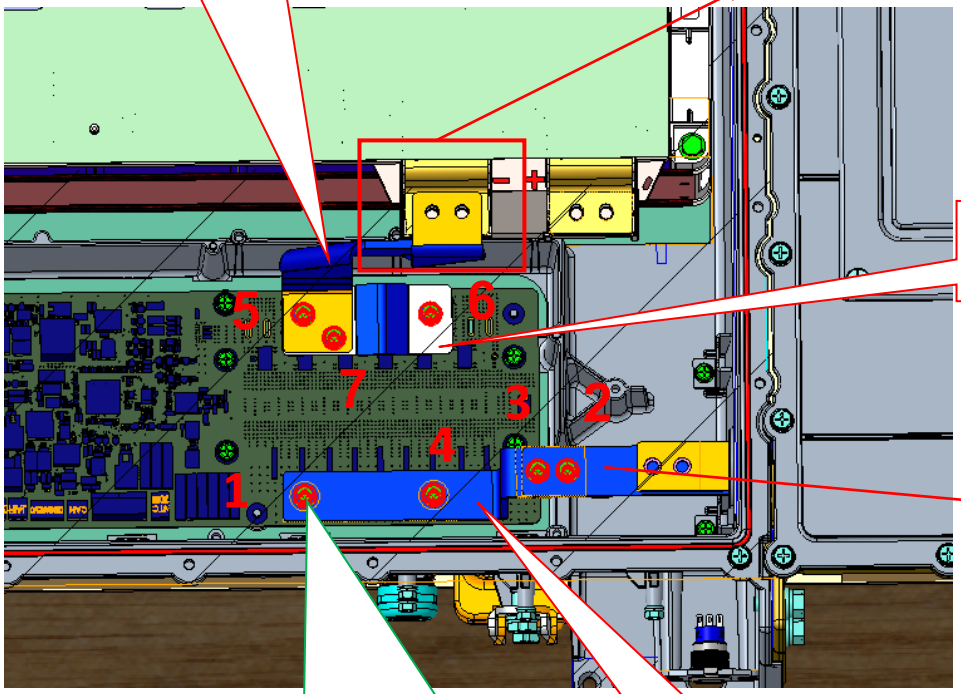
NO.	Code	Description	Quantity
1			
2			
3			
4			

Process 7:Install positive and negative copper bars

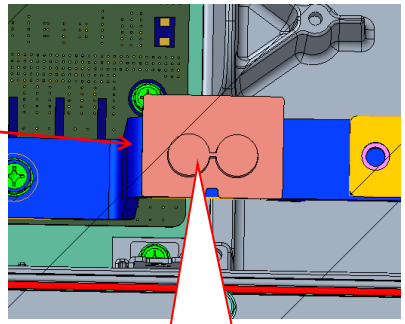


Make sure that the copper bar is aligned with the thread hole of the module before tightening the 3pcs screws at the bottom of the negative copper

25020010 Module negative connection flexible copper bar



25020009 BMS negative connection flexible copper bar



26010091 M6x16 combination Screw-coated with colored zinc 7pcs;Tightening torque 8N. m

25010147 Negative copper bar

25060106 Silicone fixing sleeve

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	26010091	M6x16 combination Screw-coated with colored zinc	7
2	25010147	Negative copper bar	1
3	25020009	BMS negative connection flexible copper bar	1
4	25020010	Module negative connection flexible copper bar	1
5	25060106	Silicone fixing sleeve	1

Procedure of operation

Step 1:Use a scannner scan the barcode of the semi-finished product & AM7 & AU8

Step 2:Place the Negative copper bar on AU8 and tighten the 4pcs screws;

Step 3:Install the silicone fixing sleeve onto the top screw of the negative copper bar;

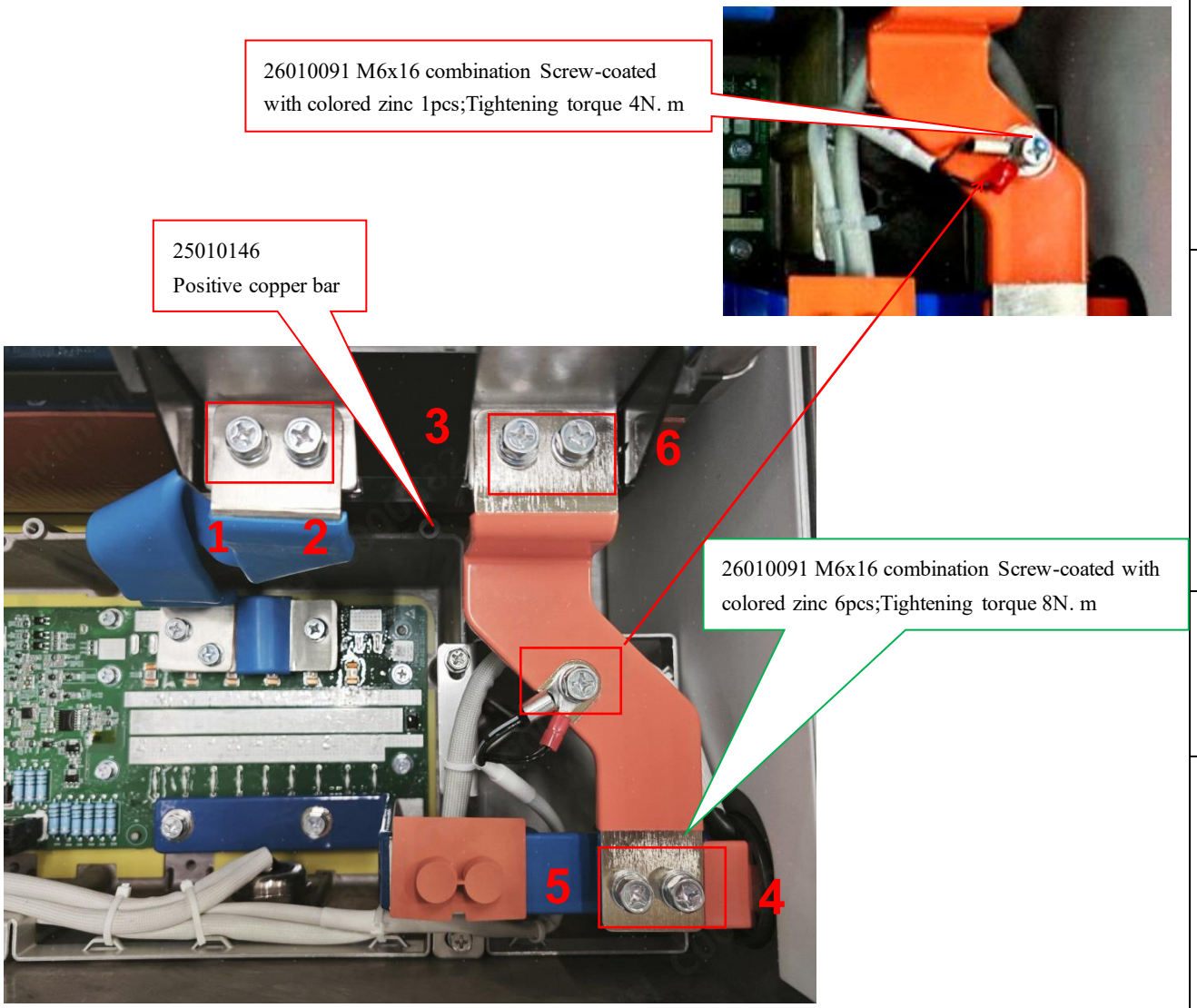
Step 4:Place the negative connection flexible copper bars and tighen the 3pcs screws at the bottom;

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Intelligent electric screwdriver	1
2			
3			
4			
5			

Process7: Install positive and negative copper bar



Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	26010091	M6x16 combination Screw-coated with colored zinc	7
2	25010146	Positive copper bar	1
3			
4			

Procedure of operation

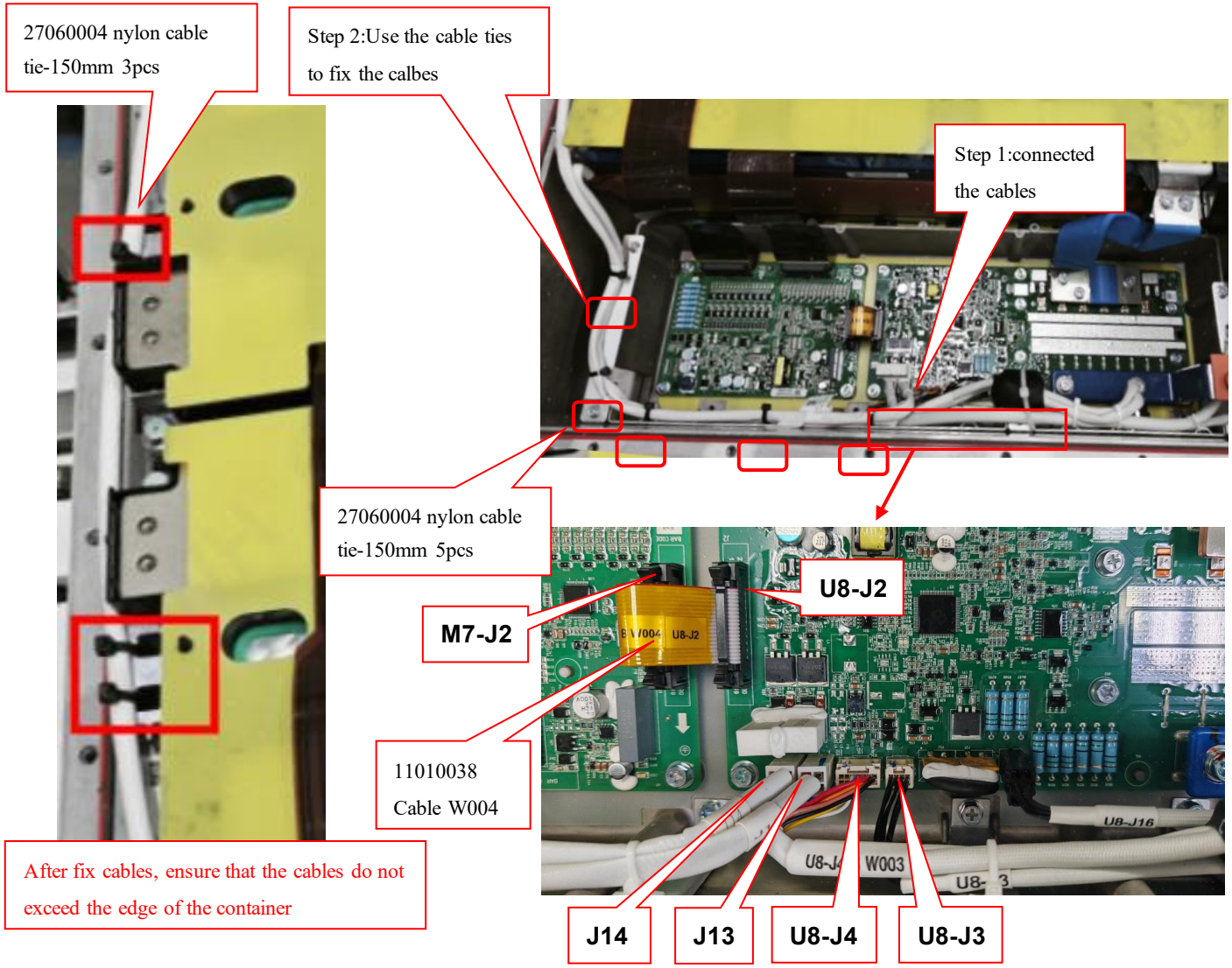
Step 5:Tighten the 2pcs screws at the top of module negative connection flexible copper bar;
Step 6:Place the Positive copper bar on the module and tighten the 4pcs screws;
Step 7:Use a torque wrench to fasten the OT terminal of the power cable to the copper bar ;

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Intelligent electric screwdriver	1
2	/	Torque wrench	1
3			
4			
5			

Process 8:Cable connection and fix



After fix cables, ensure that the cables do not exceed the edge of the container

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	11010038	Cable W004	1
2	27060004	150mm nylon cable tie	8
3			
4			
5			
6			

Procedure of operation

Step 1:Connect cables according to the PCBA connector number ;

order of connection：

J14,J13,M7-J2,U8-J2,U8-J4,U8-J3

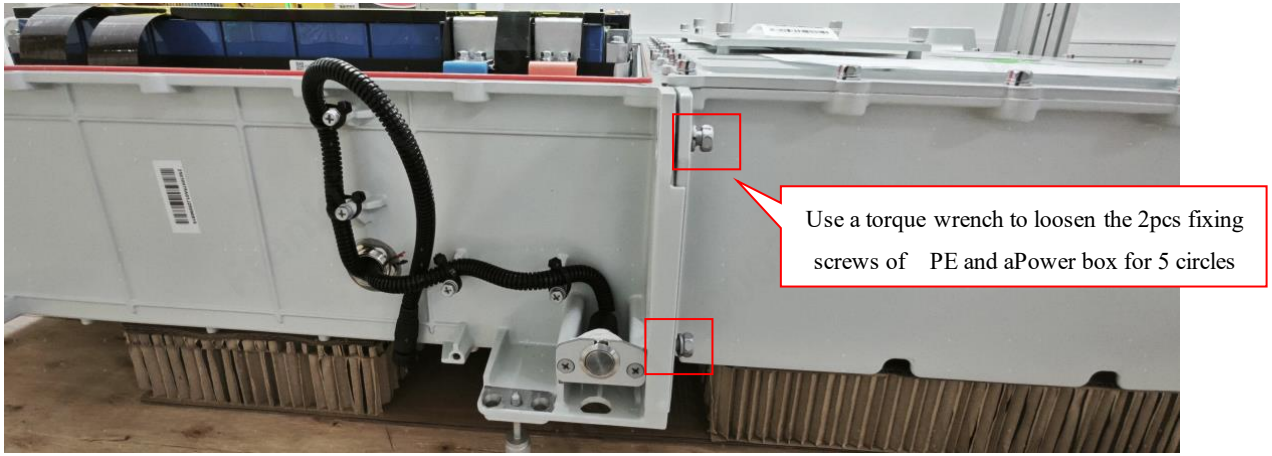
Step 2:Use the cable ties to fix the cables;

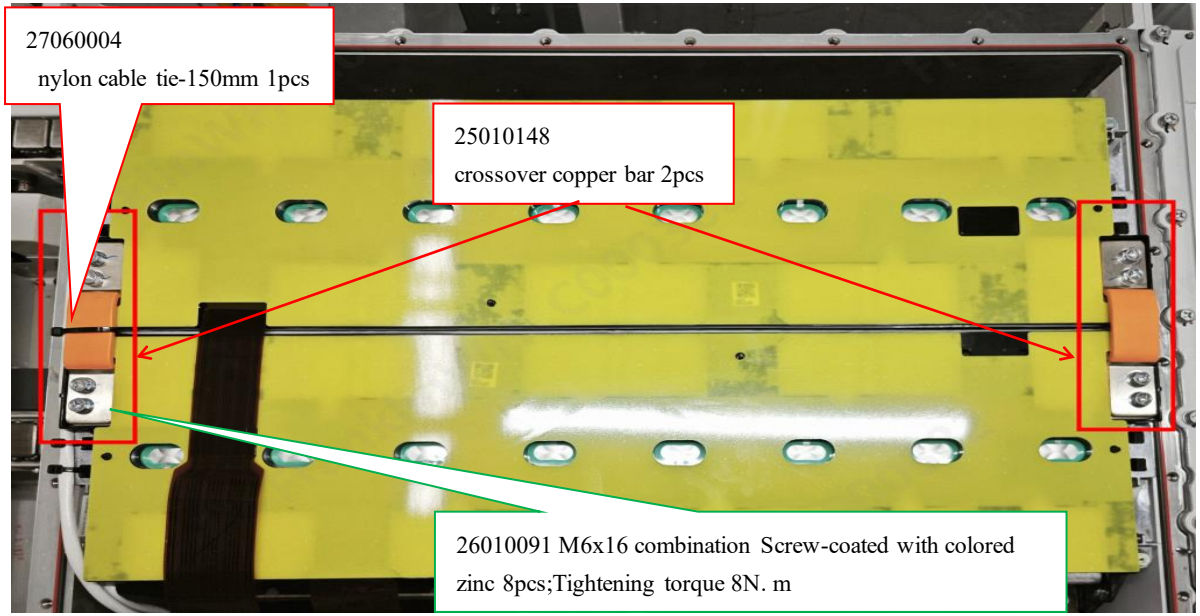
Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Diagonal pliers	1
2			
3			
4			
5			

Process 9:Install crossover copper bars





Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	25010148	crossover copper bar	2
2	26010091	M6x16 combination Screw-coated with colored zinc	8
3	27060004	150mm nylon cable tie	1
4			
5			
6			

Procedure of operation

Step 1:Use a scannner scan the barcode of the semi-finished product

Step 2:Use a torque wrench to loosen the 2pcs fixing screws of PE and aPower box for 5 circles;

Step 3:Place the crossover copper bar on the battery module and tighten the screws;

Step 4:Place the crossover copper bar on the battery module and tighten the screws;

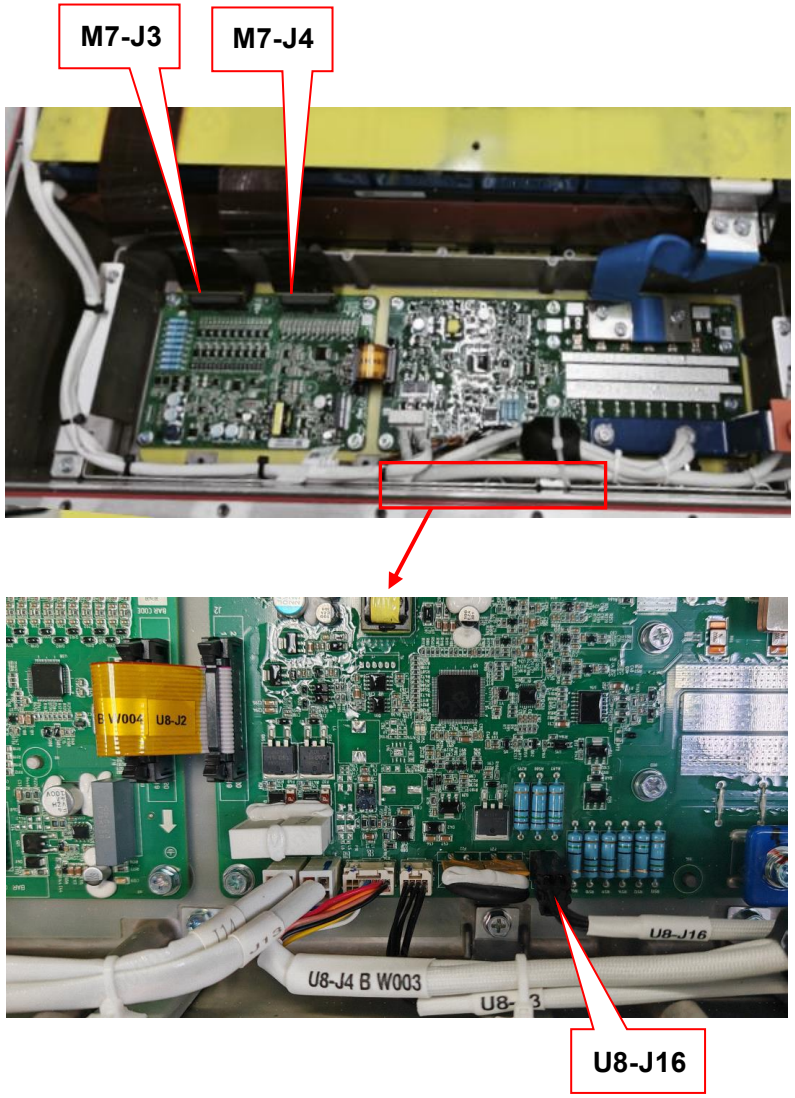
Step 5:Use the cable ties to fix the cable to crossover bar

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Torque wrench	1
2			
3			
4			
5			

Process 9:Install crossover copper bars



Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	25010148	crossover copper bar	2
2	26010091	M6x16 combination Screw-coated with colored zinc	8
3	27060004	150mm nylon cable tie	1
4			
5			
6			

Procedure of operation

Step 6:Connect cables according to the PCBA connector number ;

order of connection：

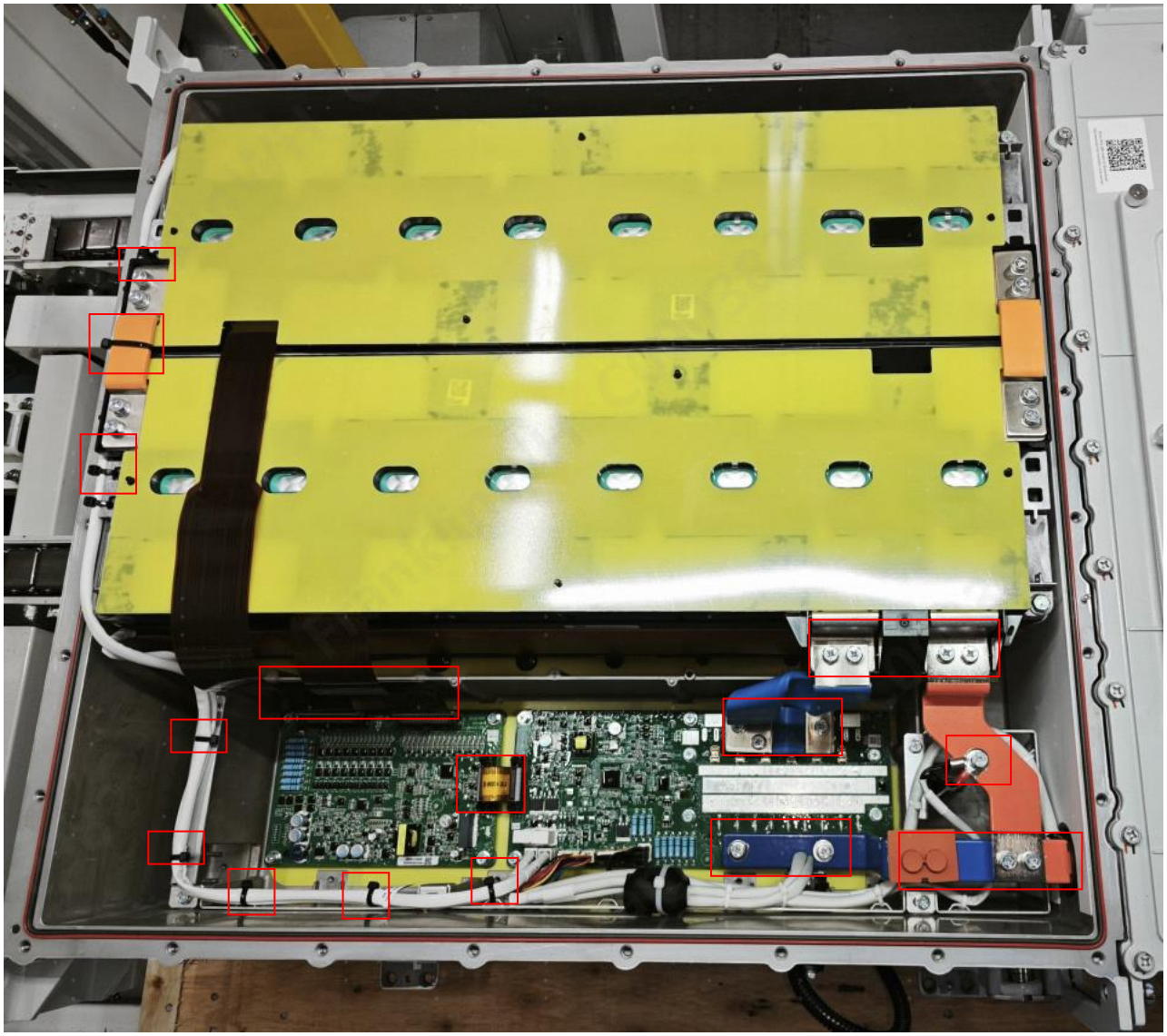
M7-J3,M7-J4,U8-J16

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Torque wrench	1
2			
3			
4			
5			

Process 10:FQC inspection



Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Use Torque wrench to check torque of Screw on positive and negative copper bars

Step 2: Check if the cable install properly.

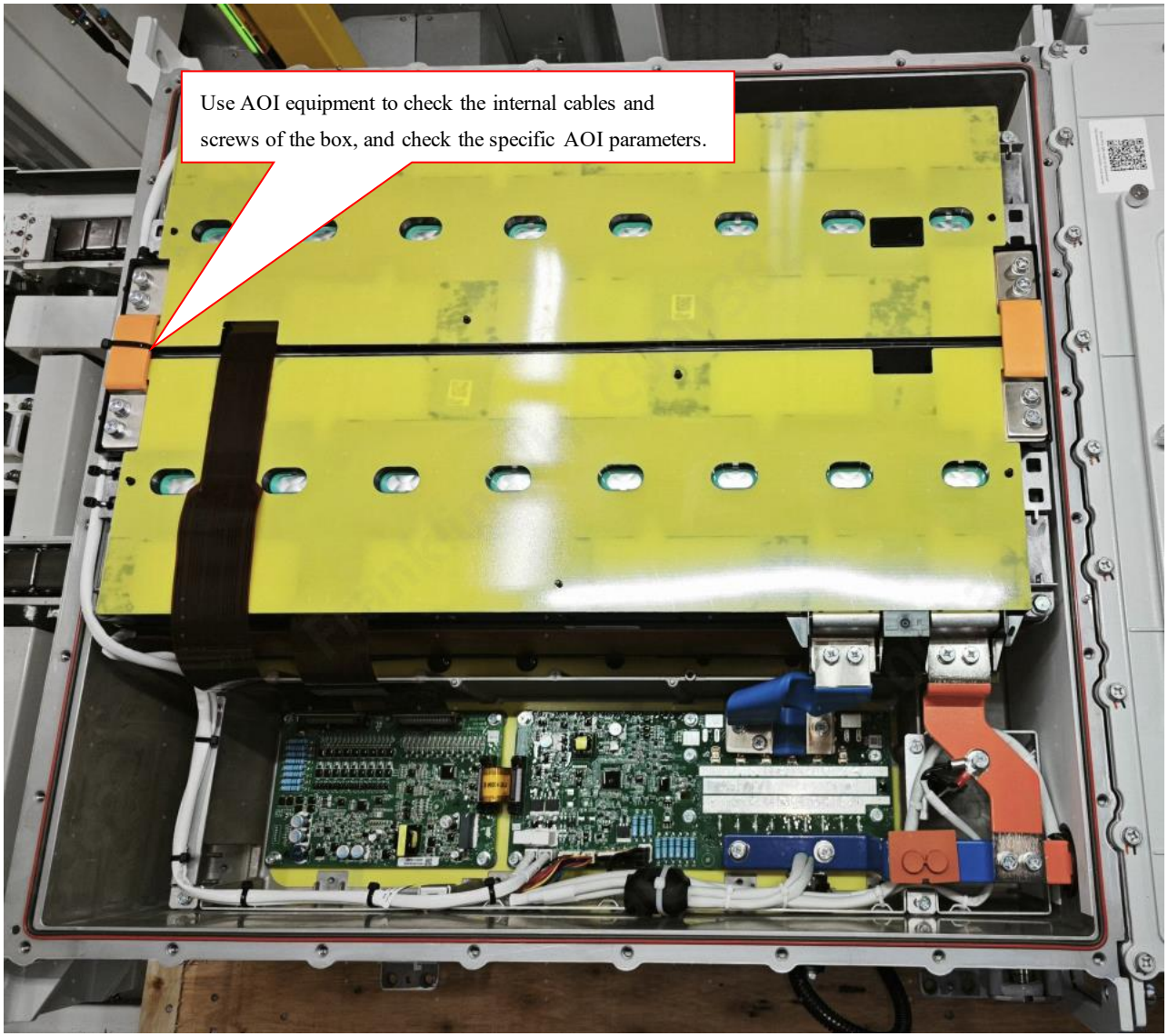
Step 3: Check if the cable fixed properly.

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Torque wrench	1
2			
3			
4			

Process 11:AOI Inspection



Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Use AOI equipment to check the internal cables and screws of the box, and check the specific AOI parameters.

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	AOI equipment	1
2			
3			
4			
5			

Process 12:Install the aPower front cover

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	25100022	Front cover	1
2	26010181	M6x16 combination Screw-coated with dacromet	22
3			
4			

Procedure of operation

Step 1:Use an assistive arm to move the aPower front cover to the specified position,and Tighten 22pcs screws;

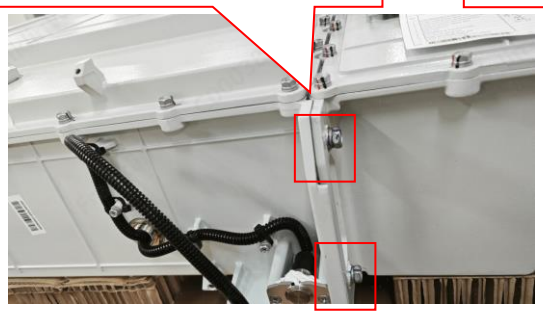
Notes

Tool and equipment

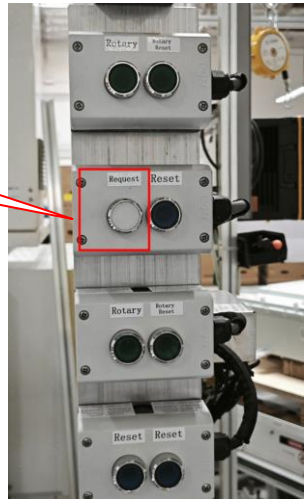
NO.	Code	Description	Quantity
1	/	Assistive arm	1
2	/	Automatic screw tightening equipment	1
3			
4			
5			

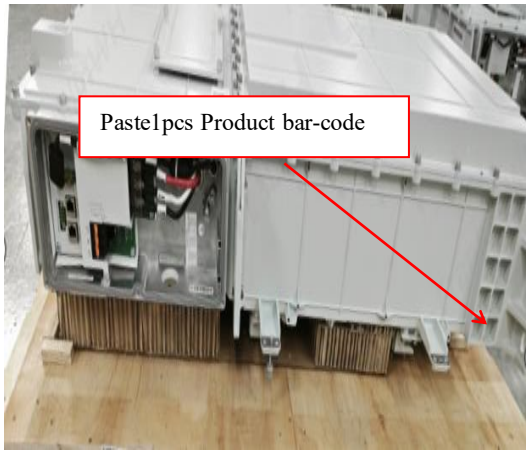
Process 13:Safety testing

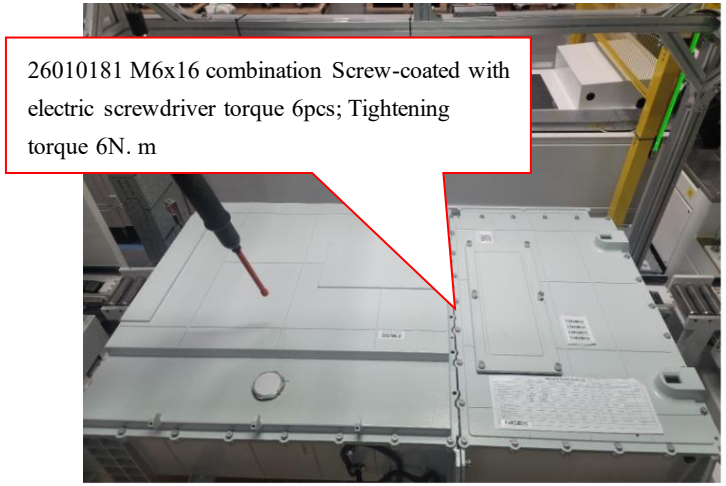
Use a torque wrench to tighten the 2pcs fixing screws of PE and aPower box, and tighten torque 15N.m



Press the white button shown in the diagram to rotate the product 180 degrees horizontally.







Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	26010181	M6x16 combination Screw-coated with electric screwdriver torque	6
2	/	Torque wrench	1
3			
4			
5			
6			

Procedure of operation

Step 1: Scan semi-production S/N and product S/N

Step 2: Use a torque wrench to tighten the 2pcs fixing screws of PE and aPower box and tighten torque 15N.m.

Step3: Manually use an electric screwdriver torque to tighten the 6pcs and tighten torque 6N.m

Step 4: Press the white button shown in the diagram to rotate the product 180 degrees horizontally.

Step 5: paste 1pcs S/N label on the user interface window

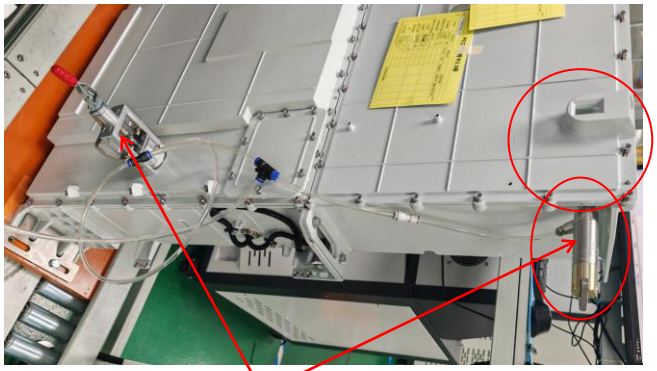
Step6: Safety test according to Testing instructions.

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Torque wrench	1
2			
3			
4			
5			
5			

Process 14:Leak&BMS&aPower functional test



Step 1:Connect PACK and PE module leak joint

Step 2:Connect the AC test cables and the network cable to the aPower

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Connect PACK and PE module leak joint;

Step 2:Connect the AC test cables and the network cable to the aPower;

Step 3:Operate the equipment for testing;

Step 4:After functional testing, transport offline to the aging testing area for aging testing;

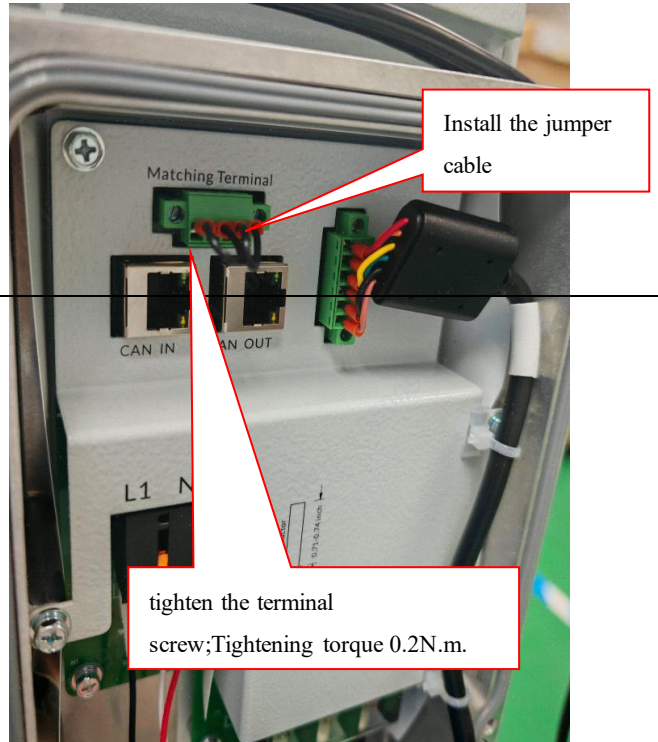
	Notes	1.Leak&BMS&aPower function test For details, see the debugging instructions;			
	Tool and equipment	NO.	Code	Description	Quantity
		1			
		2			
		3			
		4			
		5			
Refer to Function aging testing instruction	**Operating instruction**				
	Bill of materials	NO.	Code	Description	Quantity
		1			
		2			
		3			
		4			
		5			
		6			
	Procedure of operation				

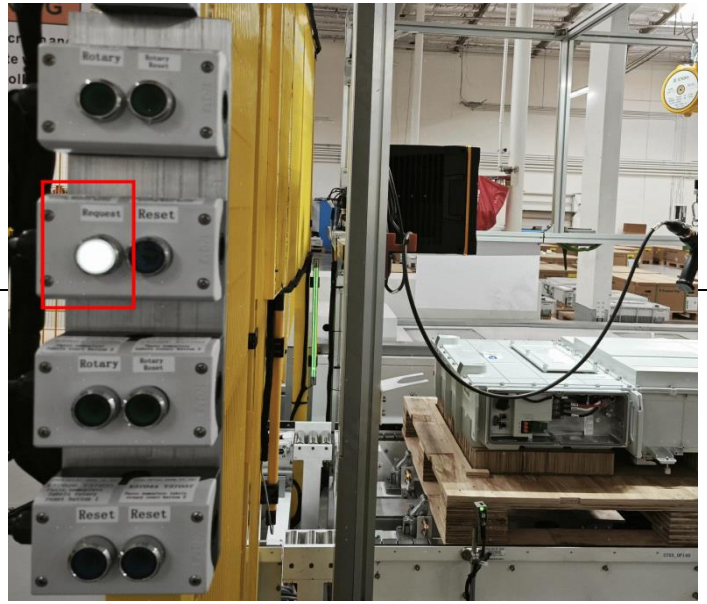
Notes

Tool and equipment

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			

Process 15:Paste nameplate label and install Breathable valve





Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	11040024	Jumper cable	1
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Install the jumper cable and tighten the terminal screw;Tightening torque 0.2N.m.

Step 2: Press the white button shown in the diagram to rotate the product 180 degrees horizontally.

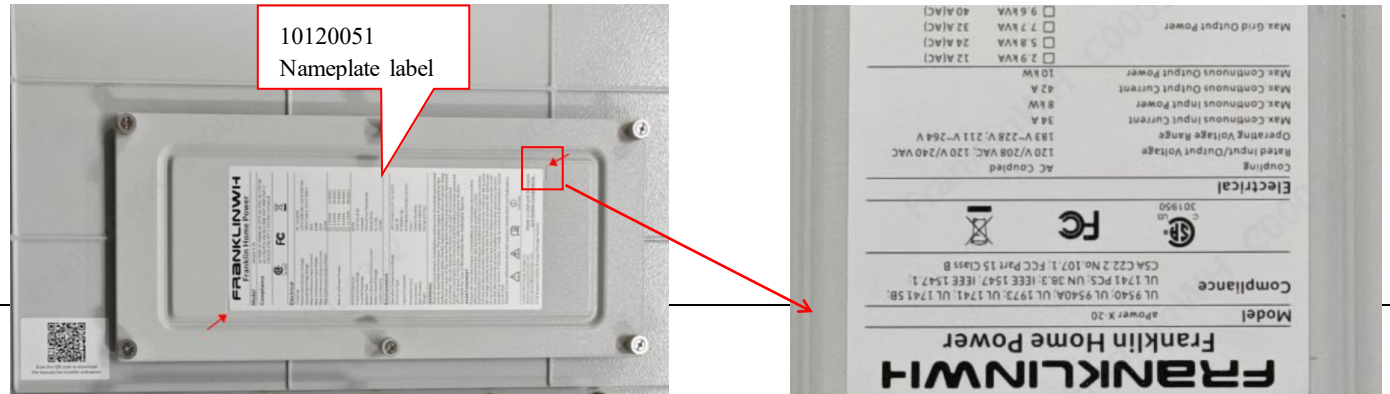
Step3:paste the barcode on the pallet as shown in the diagram.

Notes

Tool and equipment

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			

Step 16:Paste nameplate label and install Breathable valve



Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	10120051	Nameplate label	1
2	25080003	Breathable valve	1
3			
4			
5			
6			

Procedure of operation

Step 4:Scan the Semi-product S/N and product S/N

Step 5:Stick the nameplate label according to the diagonal steel stamp;

Step 6: lock the vent valve, tightening torque 0.6N. m

Step 7:Remove the process flow card

Step 8:

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	Screw driver	1
2			
3			
4			
5			

Process 17: aPower Packing

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	27060061	PET-Packing strap	5.4m
2			
3			
4			
5			
6			

Step 1: place the upper cover of the carton on the aPower

+

Procedure of operation

Step 1: place the upper cover of the carton on the aPower;

Step 2:Use automatic packing equipment to tie two straps in the transverse direction

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	automatic strapping machine	1
2			
3			
4			
5			

Process 18:aPower Packing

Step 1:Stack the two aPower units together using an automatic lifting equipment

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Stack the two aPower units together using an automatic lifting equipment;

Step 2:Use automatic packing equipment to tie two straps in the transverse direction;

Notes

Tool and equipment

NO.	Code	Description	Quantity
1	/	automatic strapping machine	1
2	/	robotic arm	1
3			
4			
5			

Process 19:Paste barcode and label

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	10120050	Class 9 Label	2
2	27060020	Adhesive printing paper	1
3			
4			
5			

	6			
Procedure of operation	Step 1:Use adhesive paper to print the shipping mark and paste it on the picture frame of the carton; Step 2:Paste the finished barcode at the S/N position; Step 3:Paste 2pcs class 9 label inside square boxes on both sides facing each other;			
Notes	Check the SKU#: APR-10K15V2U-US Check the MODEL: aPower X-20			
Tool and equipment	NO.	Code	Description	Quantity
	1			
	2			
	3			
	4			
	5			

Process 19:Paste barcode and label

Operating instruction

Bill of materials

NO.	Code	Description	Quantity
1	10120054	Origin Label	2
2			
3			
4			
5			
6			

Procedure of operation

Step 1:Paste 2pcs origin label inside square boxes on both sides

Notes

Tool and equipment

NO.	Code	Description	Quantity
1			
2			
3			
4			
5			